Practical No 08

A) Create a PWA Application with an HTML page that would be the starting point of the application. This HTML will contain a link to the file named manifest.json

Project Name: BasicPWA

Project Type: ASP.NET Core Web App (Model-View-Controller)

Program Code:

I. Filename: Program.cs

```
var builder = WebApplication.CreateBuilder(args);

var app = builder.Build();

app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

II. Filename: index.html

```
<!DOCTYPE html>
<html>
<head>
<title>Basic PWA</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="manifest" href="manifest.json">
</head>
<body>
<h1>My Progressive Web Application</h1>
<p>This is the starting page of PWA.</p>
</body>
</html>
```

III. Filename: Manifest.json

```
{
"name": "Basic Progressive Web Application",
"short_name": "BasicPWA",
"description": "Simple PWA Example",
"start_url": "/index.html",
"display": "standalone",
"background_color": "#ffffff",
"theme_color": "#0d6efd",
"orientation": "portrait",
"icons": [
{
"src": "icons/icon3.png",
"sizes": "128x128",
"type": "image/png"
},
{
"src": "icons/icon4.png",
"sizes": "128x128",
"type": "image/png"
}
]
}
```

B)Create a simple Web API that returns student data in JSON format.

Project Name: StudentAPI

Project Type: ASP.NET Core Web API`

Program Code:

I.Filename: Student.cs

```
namespace StudentAPI.Models
{
    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Course { get; set; }
        public int Age { get; set; }
    }
}
```

II.Filename: StudentsController.cs

```
using Microsoft.AspNetCore.Http;
using Microsoft.AspNetCore.Mvc;
using StudentAPI.Models;

namespace StudentAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentsController : ControllerBase
    {
        [HttpGet]
        public List<Students> GetStudents()
        {
            return new List<Students>()
            {
                new Students
                {
                    Id = 1,
                    Name = "Rahul",
                    Course = "BSc IT",
                    Age = 20
                },
                new Students
                {
                    Id = 2,
                    Name = "Sneha",
                    Course = "BCA",
                    Age = 21
                }
            };
        }
    }
```